Potla Naga Sai Prajith

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.97
Sri Chaitanya Junior College, Pratibha Campus	Class XII (TSBIE)	2023	98.30%
Sri Chaitanya Techno School, Kukatpally	Class X (SSC)	2021	10.00%

WORK EXPERIENCE

Agentic AI Intern, Oracle

Tech: Oracle AI Agent Studio, Fusion REST APIs, Agentic Workflows

- Developed an **Agentic AI application** with **3 agentic workflows** in **Oracle AI Agent Studio** to detect and resolve project-financial risks, self-identifying **2 customer-focused use cases** from an open-ended business brief and validating them with **automated schema checks**.
- Integrated **Oracle Fusion REST APIs** through **LLM and Business Object nodes**, constructing schema-driven **JSON payloads** for structured enterprise-data retrieval across all workflows with **~80 nodes.**
- Implemented **human-in-the-loop** controls for actions with **SWITCH**-based routing to dismiss malformed action payloads before execution, ensuring a **recommend-before-act boundary** where all data updates require explicit user confirmation via **REST-backed** actions.

PROJECTS

Lossless Data Compression using String Matching

DSA Lab Course Project, Prof. Vikas Krishnamurty

- Implemented *efficient* classical string matching algorithms such as **KMP, Suffix Trees, Finite Automata** for pattern recognition in large texts.
- Built a **lossless text-compression system** by integrating suffix-tree-based substring matching with **LZ-style literals and tokens**, using a minimum match length of **4**.
- Validated end-to-end losslessness across compression and decompression cycles using **byte-level comparision** and ***SHA-256*** verification.

Credit Card Fraud Detection

Group Project

- Analyzed a highly imbalanced Kaggle dataset containing **~2.84 lakh transactions and 30 features**, performing exploratory analysis and preprocessing; removed the **Time** feature based on its limited variation between fraud and non-fraud transactions.
- Evaluated Logistic Regression, **XGBoost Classifier,** and Neural Network models for binary fraud classification using an **80:20 train-test split**, with precision, **recall, F1-score, and Precision-Recall** curves as evaluation metrics.
- Applied **SMOTE** on the training data to address class imbalance and evaluated its impact on the **XGBoost Classifier**, improving test-set **recall from 0.755 to 0.906**
- Compared model performance using **AUC-PR**, achieving **0.76** with Logistic Regression and approximately **0.85** with the Neural Network on the test set

Fruits and Vegetables Disease Detection

Personal Project

- Built a **CNN model in PyTorch** to classify images of fruits and vegetables by both **name** (e.g., apple, tomato) and **condition** (fresh or rotten) using **~29,400** labeled images from Kaggle (28 classes).
- Preprocessed image data through **resizing**, **tensor conversion** and trained the model using **Adam optimizer** with **cross-entropy loss** function and achieved **97% training accuracy** and **84% test accuracy** in predicting object type and its health.
- Currently working towards using **class weighting** to handle class imbalance and building a **two-stage pipeline** for

separate object and health classification.

SKILLS

Languages & Tools : C, C++, Python, SQL, Latex, HTML, CSS

Libraries & Frameworks: Numpy, Pandas, Matplotlib, Scikit-learn, XGBoost, PyTorch

RELEVANT COURSES

Mathematics: Probability , Applied Statistics, Numerical Linear Algebra, Convex Optimization, Multi-Agent Systems, Spectral Graph Theory

Computer Science: DBMS, Data Structures & Applications, Algorithms, Theory of Computation, Operating Systems

SCHOLASTIC ACHIEVEMENTS

- One of the 317 candidates who qualified **PRMO 2019**, Telangana
- Solved **400+ problems** on Leetcode (Profile)
- Awarded **Gold Medal** at the school level in **SOF - International Math Olympiad**
- Awarded **Reliance Foundation Scholarship** in 2023

POSITIONS OF RESPONSIBILITY

- Managed communication and facilitation for a group of students from **IIT Hyderabad** to participate in a research project under the mentorship of **Prof. Vikram Bansal** at **IIT Bombay**
- **Proctored for internship and placement recruitment tests** organized by **OCS, IIT Hyderabad**, ensuring smooth and fair execution of examination procedures.

EXTRACURRICULAR

- Solve mathematically intensive problems involving programming-based computations on the **Erdos** platform.
- Studying Philosophical books and ancient literature like **BhagavadGita**, **Ramayana** and **Mahabharata**